STAT 441 | PROJECT 2

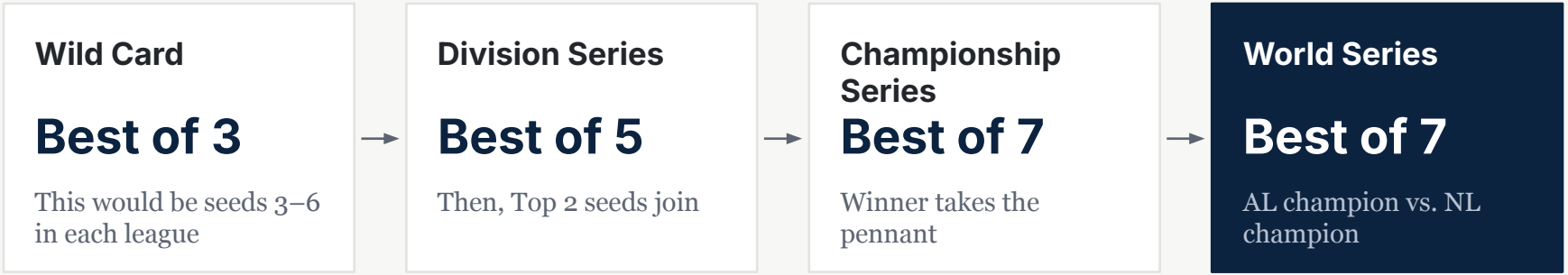
Simulating the 2026 MLB Postseason with a Markov Chain

How well does regular-season play predict who wins in October?

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Twelve teams reach October, and short series decide who advances



- A best-of-3 or a best-of-5 can turn on one bad game, so the better team does not always advance
- Seeding favors division winners, so it doesn't always match team strength.

We asked how well regular-season numbers predict the odds of winning a title

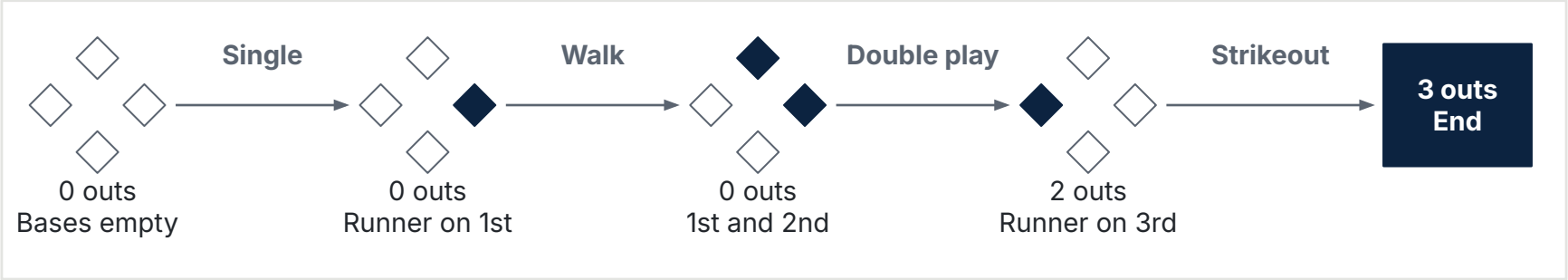
Why it matters

- Teams, fans and betting markets all want odds before October
- Series odds show which upsets were truly surprising
- 2026 gives a real test: 11 series with known winners

Limits of simple approaches

- A record compresses a whole season into one number
- It ignores how a lineup fares against a specific pitching staff
- It ignores how runs are built: runners, outs and timing

Step 1: Model each half-inning as moves between 24 base-out states



24 states + 1 end state

3 out counts × 8 runner combinations; the chain stops at 3 outs.

Why states

A single scores a runner from 2nd but not with the bases empty, so runs depend on the situation.

Data

Every 2026 regular-season plate appearance from Statcast, innings 1–8.

Step 2: Combine each team's hitting with its opponent's pitching

01

Sort plate appearances

8 outcomes: out, strikeout, walk/HBP, single, double, triple, home run, error.

Why: these are the moves that change the base-out state.

02

Shrink team rates

Add 200 league-average plate appearances to each team's record.

Why: small samples would otherwise swing the estimates.

03

Pair offense with pitching

Combine the batting team's rates with the pitching team's using log5 (odds ratio).

Why: gives outcome rates for one specific matchup.

Note: how runners move after each outcome comes from league-wide 2026 data, so every team's chain shares the same baserunning.

ANALYSIS

Step 3: Turn half-innings into a single-game win probability



Exact, not simulated

The full distribution of runs per half-inning is computed directly, so game odds carry no simulation noise.

Both sides of the matchup

Each team bats against the other's pitching; home and away are computed separately.

Verified

The exact answer matched 10,000 simulated games (Dodgers vs. Brewers).

ANALYSIS

Step 4: Play the real 2026 bracket 10,000 times

01 Game
Win probability for each home/away pairing

02 Series
Play games until one team has enough wins

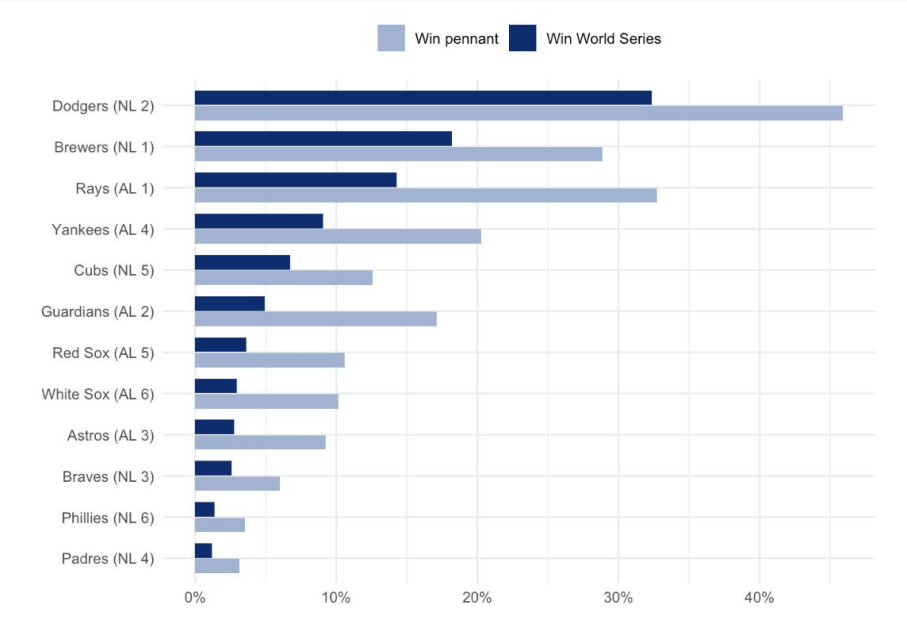
03 Bracket
Every round, from the real 2026 seeds

04 10,000 runs
Count pennants and titles for each team

Round	Length	Home games
Wild Card	Best of 3	All at higher seed
Division Series	Best of 5	2-2-1
Championship Series	Best of 7	2-3-2
World Series	Best of 7	2-3-2, more wins hosts

Then score the model: its probability for each of the 11 real 2026 series vs. who actually won.

The Dodgers led title odds at 32%, nearly double the top-seeded Brewers



Rays led the AL

32.7% pennant, 14.3% title

Seeds didn't set the order

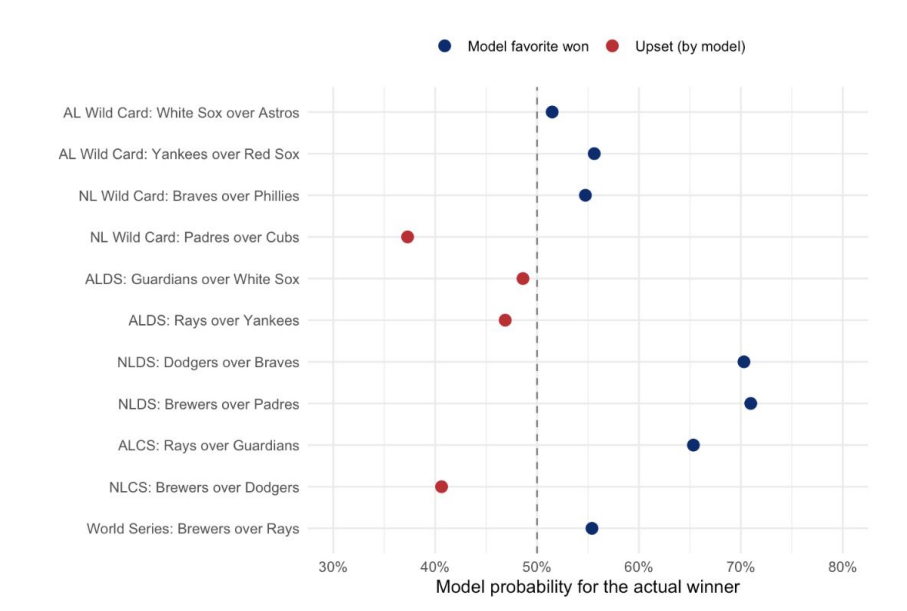
The 5-seed Cubs (6.7%) rated above the 3-seed Braves (2.6%)

A wide-open field

No team topped a 1-in-3 title chance

RESULTS

The model's favorite won 7 of 11 series in the 2026 postseason



Wild Card: 3 of 4

Favorites won three of the four best-of-3 series

Division Series: 2 of 4

Narrow favorites Yankees and White Sox both lost

Mostly close calls

In 8 of 11 series the favorite was under 63%

RESULTS

The Brewers won it all despite being a 40.6% underdog in the NLCS

71.0%
NLDS vs. Padres
Won as favorite

40.6%
NLCS vs. Dodgers
Won as underdog

55.4%
World Series vs. Rays
Won as slight favorite

Pre-playoff title odds



Title odds are not one series' odds. The 1-seed Brewers' path ran through the Dodgers, the model's title favorite, so their pre-playoff title odds were only 18.2%.

Conclusion and next steps

What We Found

- A team-level Markov chain turns regular-season data into game, series and title odds
- It favored the Dodgers (32.3%) and picked the winner in 7 of 11 real series
- Short series stay near coin flips, so upsets like the Brewers' NLCS win are common

Future Tweaks

01 Model lineups and bullpens

Use individual hitters and pitchers instead of team averages

02 Add parks and late innings

Include park effects and 9th-inning and extra-inning play

03 Test on more postseasons

11 series is too few to judge how well-calibrated the odds are